Project: Worked examples
Title: Mechanism kinematics
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1. Scope and method

Calculates the position, velocity, acceleration, input torque and forces of a one-degree-of-freedom planar four-bar, slider-crank or cam mechanism. The input turns at constant speed, and the links are rigid, with no clearance and no friction.

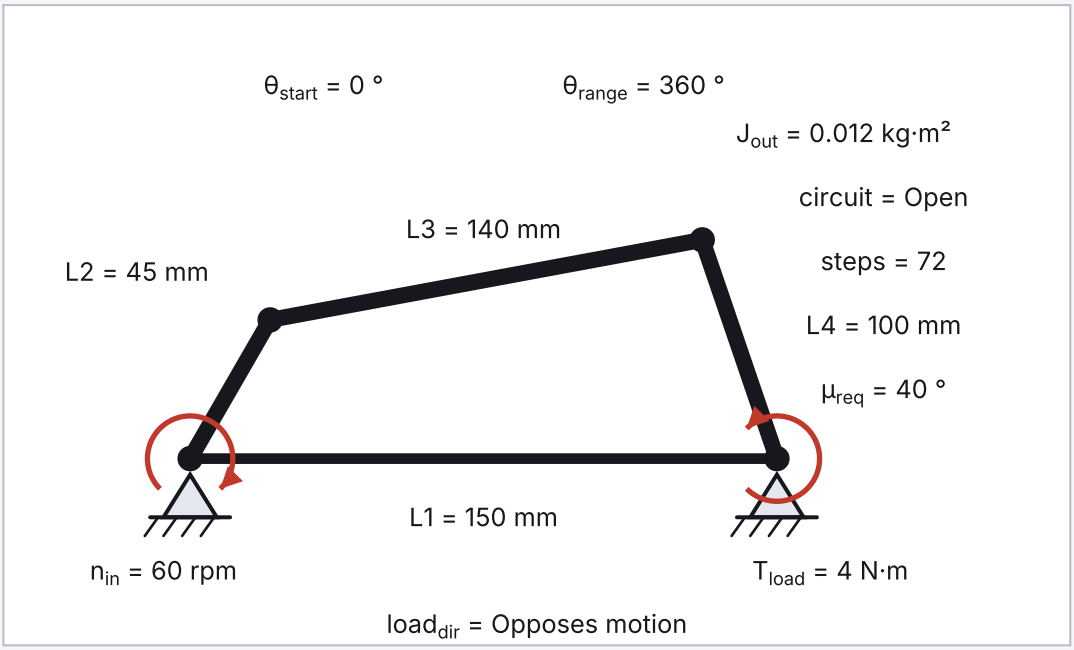


Figure 1 — A four-bar linkage: ground L1 between O_2 and O_4, crank L2 turning at n_{in}, coupler L3, rocker L4 carrying the load torque T_{load}. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Four-bar position	$a\cdot e^{j\theta_2} + b\cdot e^{j\theta_3} = d + c\cdot e^{j\theta_4}$, with $\ell = \lvert A - O_4\rvert$, $\cos\gamma = (c^2 + \ell^2 - b^2)/(2c\ell)$ and $\theta_4 = \varphi \mp \gamma$ (φ the direction $O_4 \to A$; open: −, crossed: +); θ_3 from B − A
Four-bar velocity and acceleration	differentiated loop, 2×2 system with $\det = b\cdot c\cdot\sin(\theta_3 - \theta_4)$; $\omega_3 = \omega_2\cdot a\cdot\sin(\theta_4 - \theta_2)/(b\cdot\sin(\theta_3 - \theta_4))$, $\omega_4 = \omega_2\cdot a\cdot\sin(\theta_3 - \theta_2)/(c\cdot\sin(\theta_3 - \theta_4))$; α from the second derivative at constant ω_2
Grashof class and transmission angle	$s + l < p + q$ (crank-rocker, double-crank or double-rocker by the shortest link), = at the change point, > non-Grashof; transmission angle $\cos\mu = (b^2 + c^2 - \ell^2)/(2bc)$ folded to 0 … 90°; toggle where $\ell = b + c$ or $\lvert b - c\rvert$
Slider-crank position	$x = r\cdot\cos\theta + \sigma\cdot\sqrt{(l^2 - (r\cdot\sin\theta - e)^2)}$ (σ = +1 open, −1 crossed); $dx/d\theta = -r\cdot\sin\theta - r\cdot\cos\theta\cdot(r\cdot\sin\theta - e)/(x - r\cdot\cos\theta)$; dead centres where crank and rod are collinear; $\mu = 90° - \lvert$rod angle to the slide line$\rvert$

WHAT IT CALCULATES	RELATION
Input torque and forces	$T_{in}\cdot\omega_{in} = \Sigma F\cdot v$ by virtual work; four-bar $T_{in} = (T_L + J\cdot\alpha_4)\cdot\omega_4/\omega_2$, tangential pin force $P = (T_L + J\cdot\alpha_4)/c$ and coupler force $P/\sin\mu$; slider $T_{in} = (m\cdot a + F_L)\cdot dx/d\theta$, slide-direction force $F_x = m\cdot a + F_L$, rod force $F_x/\cos\theta_3$ and guide reaction $F_x\cdot\tan\theta_3$; RMS torque and mean power over the sweep by Simpson's rule
Assembly limits and loop closure	four-bar toggles at $\cos\theta_2 = (d^2 + a^2 - \ell^2)/(2ad)$, $\ell = b + c$ or $\lvert b - c\rvert$; slider $\sin\theta = (e \pm l)/r$; the sweep is cut at them and the unbounded peaks are not reported; closure $\lvert a\cdot e^{j\theta_2} + b\cdot e^{j\theta_3} - c\cdot e^{j\theta_4} - d\rvert$ (slider: $\lvert r\cdot e^{j\theta} + l\cdot e^{j\theta_3} - (x + j\cdot e)\rvert$) below 1e-9 mm at every step
Cam motion	per segment with $x = (\theta - \theta_o)/\beta$, rise $s = s_o + h\cdot f(x)$, return $s = s_o - h\cdot f(x)$; cycloidal $f = x - \sin(2\pi x)/(2\pi)$; 3-4-5 $f = 10x^3 - 15x^4 + 6x^5$; 4-5-6-7 $f = 35x^4 - 84x^5 + 70x^6 - 20x^7$; modified trapezoid from its acceleration pieces ($C_v = 2$, $C_a = 8\pi/(\pi + 2)$, $C_j = 32\pi^2/(\pi + 2)$); SVAJ at constant ω: $v = \omega\cdot s'$, $a = \omega^2\cdot s''$, $j = \omega^3\cdot s'''$ ($' = d/d\theta$), so $v_{pk} = C_v\cdot h\cdot\omega/\beta$
Cam contact	roller pressure angle $\varphi = \arctan[(s' - e)/(\sqrt{(R_p^2 - e^2)} + s)]$ with $R_p = R_b + R_r$, base circle $R_{p,req} = \sqrt{(s_o,req^2 + e^2)}$ with $s_o,req = \max(\lvert s' - e\rvert/\tan\varphi_{max} - s)$; pitch-curve radius $\rho = \lvert P'\rvert^3/(P' \times P'')$, undercut where $\rho_{min} \le R_r$; flat-faced profile radius $\rho = R_b + s + s''$ above zero; follower force $F = m\cdot a + k\cdot s + F_{pre} + F_{ext}$ (the follower leaves the cam where $F < 0$), input torque $T = F\cdot s'$, roller normal force $F/\cos\varphi$

Basis: Position, velocity and acceleration analysis; transmission angle; Grashof condition[1](Ch. 4, 6, 7, 6th ed.); Loop-closure position analysis; velocity and acceleration of planar linkages[2](Ch. 4–6, 5th ed.).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
mode	Kind of mechanism	Four-bar		spec: Carton line PKG-LINE-7 §3.2 tucker station rev C	any
circuit	Assembly circuit (four-bar, slider-crank): open is $\theta_4 = \varphi - \gamma$ (slider on the +x side of the crank pin), crossed the mirror branch	Open		drawing: PKG-TUCK-02 rev B	any
L1	Ground link d, O_2 to O_4 (four-bar)	150	mm	drawing: PKG-TUCK-02 rev B	1 … 10000 mm
L2	Crank length a (four-bar) or r (slider-crank)	45	mm	drawing: PKG-TUCK-02 rev B	1 … 10000 mm
L3	Coupler length b (four-bar) or connecting-rod length l (slider-crank)	140	mm	drawing: PKG-TUCK-02 rev B	1 … 10000 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
L4	Rocker length c (four-bar)	100	mm	drawing: PKG-TUCK-02 rev B	1 … 10000 mm
steps	Intervals N of the table (N + 1 rows, a whole number, at most 3600; the peaks do not depend on it)	72		assumption: 5° table step.	4 … 720
n_{in}	Input speed (constant)	60	rpm	spec: Carton line PKG-LINE-7 §3.2: 60 cartons/min rev C	0.1 … 30000 rpm
θ_{start}	Input angle at the first row of the table	0	°	spec: Carton line PKG-LINE-7 §3.2: crank zero at the home position rev C	-720 … 720 °
θ_{range}	Swept range of the input angle (360° for a full turn)	360	°	spec: Carton line PKG-LINE-7 §3.2: one cycle rev C	1 … 720 °
$load_{dir}$	External load: constant direction, or always opposing the output motion	Opposes motion		assumption: The flap resists the blade in whichever direction it moves.	any
T_{load}	External load torque on the output link (four-bar). Positive acts in the $-\theta_4$ direction, resisting positive ω_4; with 'Opposes motion' its sign is applied against the actual direction of ω_4 (a negative value then assists)	4	N·m	assumption: Example flap-folding torque at the rocker shaft; measure it on the station.	-10000 … 10000 N·m
J_{out}	Moment of inertia of the output link about its pivot (four-bar)	0.012	kg·m²	assumption: Example: rocker arm and blade about the rocker pivot; take it from the CAD mass properties.	0 … 1000 kg·m²
μ_{req}	Minimum transmission angle required (linkages)	40	°	assumption: Default minimum transmission angle; set it from your design rule.	10 … 80 °
follower	Follower type: roller (translating, pressure angle and undercut checked) or flat-faced (in-line)	Roller		assumption: Cam mode only; this four-bar linkage does not use it, so the template default is kept.	any

3. Assumptions

A1. The mechanism is rigid and planar, has one degree of freedom, and has no joint clearance or friction. (Scope limits of the model.)
A2. The input turns at a constant speed. (Scope limits of the model.)
A3. The position is the closed-form solution of the vector loop, on the chosen circuit. (Open and crossed are the two branches: $\theta_4 = \varphi - \gamma$ and $\varphi + \gamma$ for the four-bar, $\sigma = +1$ and -1 for the slider.)
A4. The sweep stays on the chosen circuit and follows it by continuity through a change point. (Only a change point, with all four links collinear ($\theta_2 = 0$ or π), flips the branch.)
A5. A linkage with one change point per turn repeats every 720°, and a sweep of more than one turn warns. (A kite, for example 40/40/100/100, is such a linkage.)
A6. The table leaves out angles where the linkage cannot be assembled or sits at a toggle. (It counts them, and at a toggle the velocity system is singular.)
A7. The input torque comes from the output load and the inertia of the output element only. (The crank and the coupler or connecting rod are massless: add their mass to the output element for a first allowance.)
A8. The crank, the coupler and the connecting rod are massless two-force members. (Rigid body, frictionless.)
A9. The peaks are those of the continuous motion over the swept range, not of the printed table. (A dense scan refined by golden section finds them, so they do not depend on the step count of the table (up to 3600).)
A10. At an assembly limit the sweep is cut, and the unbounded peaks, torque and forces are not reported. (The limit angles are listed: $\ell = b + c$ or $|b - c|$ (four-bar), $|r \cdot \sin\theta - e| = l$ (slider).)
A11. A change-point four-bar is continued through its singular position on the same branch. (Its speeds and torques next to the singular position are not reported.)
A12. The cam drives a translating follower at constant speed. (Norton, Design of Machinery, cam design: the four classical laws, pressure angle and radius of curvature.)
A13. The follower force is $m \cdot a + k \cdot s + F_{pre} + F_{ext}$, and contact is lost below zero. (The spring rate and the preload are those at the base circle, and the segment table must add to 360° and close.)
A14. Pressure angle and undercut are checked for the roller follower, and the profile radius for the flat-faced one. (The roller has the prime radius $R_p = R_b + R_r$.)
A15. The output speed is a kinematic proxy at constant input speed, not the speed of an actuated cylinder. (For motor sizing, use the peak input_{torque} for the peak check and $T_{rms,in}$ with n_{in} for the thermal sizing. The mean power is P_{mean}.)
A16. steps (Intervals N of the table (N + 1 rows, a whole number, at most 3600; the peaks do not depend on it)) is taken as 72. (5° table step.)
A17. load_{dir} (External load: constant direction, or always opposing the output motion) is taken as Opposes motion. (The flap resists the blade in whichever direction it moves.)
A18. T_{load} (External load torque on the output link (four-bar). Positive acts in the $-\theta_4$ direction, resisting positive ω_4; with 'Opposes motion' its sign is applied against the actual direction of ω_4 (a negative value then assists)) is taken as 4 N·m. (Example flap-folding torque at the rocker shaft; measure it on the station.)
A19. J_{out} (Moment of inertia of the output link about its pivot (four-bar)) is taken as 0.012 kg·m². (Example: rocker arm and blade about the rocker pivot; take it from the CAD mass properties.)
A20. μ_{req} (Minimum transmission angle required (linkages)) is taken as 40 °. (Default minimum transmission angle; set it from your design rule.)
A21. follower (Follower type: roller (translating, pressure angle and undercut checked) or flat-faced (in-line)) is taken as Roller. (Cam mode only; this four-bar linkage does not use it, so the template default is kept.)

4. Working

[10] $f_{in} = n_{in} / 60$
$= \text{rpm to Hz}$
$= 1\ \text{Hz}$

Input (crank) frequency, for a spring surge check

[17] $S_{plus,L}$ = s + l [3](criterion)

= shortest + longest link

= 195 mm

Shortest plus longest link

[18] $P_{plus,Q}$ = p + q [3](criterion)

= the other two links

= 240 mm

Sum of the other two links

[19] $theta2_{sw}$ = θ_2 [4](cosines))

= 73 rows, θ_2 from 0° to 360°

= [0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, … (73 items)] deg

Table: crank angle

[20] $theta3_{sw}$ = θ_3 = atan2(B - A) [4](cosines))

= 73 rows, θ_2 from 0° to 360°

= [45.45, 43.305, 41.183, 39.119, 37.143, 35.277, 33.533, 31.922, 30.446, 29.103, 27.892, 26.806, … (73 items)] deg

Table: coupler angle

[21] $theta4_{sw}$ = θ_4 = φ ∓ γ [4](cosines))

= 73 rows, θ_2 from 0° to 360°

= [93.891, 91.886, 90.182, 88.806, 87.774, 87.09, 86.749, 86.738, 87.038, 87.625, 88.475, 89.561, … (73 items)] deg

Table: rocker angle

[22] $omega3_{sw}$ = ω_3 [4](cosines))

= 73 rows, θ_2 from 0° to 360°

= [-2.6928, -2.6892, -2.6368, -2.5435, -2.4183, -2.2706, -2.1088, -1.9402, -1.7705, -1.6037, -1.4425, -1.2886, … (73 items)] rad/s

Table: coupler angular velocity

[23] $omega4_{sw}$ = ω_4 [4](cosines))

= 73 rows, θ_2 from 0° to 360°

= [-2.6928, -2.3371, -1.9398, -1.5152, -1.0781, -0.64174, -0.21757, 0.18575, 0.56219, 0.90807, 1.2216, 1.5026, … (73 items)] rad/s

Table: rocker angular velocity

[24] α_{ref} = ω_2^2 [4](cosines))

= (6.2832 rad/s)²

= 39.478 rad/s²

Angular-acceleration scale of the input (an α below 1e-9 of it is round-off and is shown as 0)

[25] $alpha3_{sw}$ = α_3 [4](cosines))

= 73 rows, θ_2 from 0° to 360°

= [-1.6439, 2.0963, 5.3499, 7.9793, 9.9336, 11.238, 11.97, 12.238, 12.157, 11.832, 11.354, 10.795, … (73 items)] rad/s²

Table: coupler angular acceleration

[26] $alpha4_{sw}$ = α_4 [4](cosines))

= 73 rows, θ_2 from 0° to 360°

= [23.794, 27.273, 29.765, 31.194, 31.596, 31.098, 29.878, 28.129, 26.035, 23.751, 21.398, 19.064, … (73 items)] rad/s²

Table: rocker angular acceleration

[27] μ_{sw} = μ [4](cosines))

= 73 rows, θ_2 from 0° to 360°

= [48.441, 48.581, 48.999, 49.687, 50.631, 51.814, 53.216, 54.816, 56.592, 58.522, 60.583, 62.755, … (73 items)] deg

Table: transmission angle

[28] $T_{in,sw} = (T_L + J \cdot \alpha_4) \cdot \omega_4 / \omega_2$ [5](work)
= 73 rows, θ_2 from 0° to 360°
= [1.5919, 1.3661, 1.1246, 0.87436, 0.62126, 0.37043, 0.1261, 0.12823, 0.38586, 0.61929, 0.82764, 1.0113, … (73 items)] N·m
Table: input torque by virtual work

[29] $theta3_0 = \theta_3$ [4](cosines))
= at the first input angle
= 45.45 deg
Coupler angle at the first input angle

[30] $theta4_0 = \theta_4$ [4](cosines))
= at the first input angle
= 93.891 deg
Rocker angle at the first input angle

[31] $omega3_0 = \omega_3$ [4](cosines))
= at the first input angle
= -2.6928 rad/s
Coupler angular velocity at the first input angle

[32] $omega4_0 = \omega_4$ [4](cosines))
= at the first input angle
= -2.6928 rad/s
Rocker angular velocity at the first input angle

[33] $alpha3_0 = \alpha_3$ [4](cosines))
= at the first input angle
= -1.6439 rad/s²
Coupler angular acceleration at the first input angle

[34] $alpha4_0 = \alpha_4$ [4](cosines))
= at the first input angle
= 23.794 rad/s²
Rocker angular acceleration at the first input angle

[35] $\mu_0 = \mu$ [4](cosines))
= at the first input angle
= 48.441 deg
Transmission angle at the first input angle

[36] $T_{in,0} = (T_L + J \cdot \alpha_4) \cdot \omega_4 / \omega_2$ [5](work)
= at the first input angle
= 1.5919 N·m
Input torque at the first input angle

[37] $theta4_{alt} = \theta_4$ on the other circuit [4](cosines))
= mirror branch, at the first input angle
= 266.11 deg
Rocker angle on the other circuit at the first input angle

[38] $theta4_{min} = \min \theta_4$ [4](cosines))
= unwrapped along the sweep
= 86.704 deg
Smallest rocker angle

[39] $theta4_{max} = \max \theta_4$ [4](cosines))
= unwrapped along the sweep
= 141.49 deg
Largest rocker angle

[40] swing = $\max \theta_4 - \min \theta_4$ [4](cosines))
= over the sweep
= 54.786 deg
Rocker swing over the sweep

[41] μ_{min} = min μ [4](cosines))
= dense scan, refined
= 48.441 deg
Smallest transmission angle over the sweep
[42] res_{max} = max |loop residual| [4](cosines))
= 73 table rows and 1441 scan angles
= below 1e-10 mm
Largest loop-closure residual
[43] $res_{vel,max}$ = max |velocity loop| / its terms [4](cosines))
= 73 table rows
= below 1e-12
Largest velocity-loop residual, relative to the size of its terms
[44] $omega3_{pk}$ = max $|\omega_3|$ [4](cosines))
= dense scan, refined
= 2.6976 rad/s
Peak coupler angular velocity
[45] $\omega_{out,pk}$ = max $|\omega_4|$ [4](cosines))
= dense scan, refined
= 3.5435 rad/s
Peak output (rocker) angular velocity
[46] $alpha3_{pk}$ = max $|\alpha_3|$ [4](cosines))
= dense scan, refined
= 22.724 rad/s²
Peak coupler angular acceleration
[47] $\alpha_{out,pk}$ = max $|\alpha_4|$ [4](cosines))
= dense scan, refined
= 31.599 rad/s²
Peak output (rocker) angular acceleration
[48] $input_{torque}$ = max $|T_{in}|$ [5](work)
= dense scan, refined
= 2.2654 N·m
Peak input (crank) torque by virtual work
[49] $output_{motion}$ = c · max $|\omega_4|$ [4](cosines))
= output pin at the rocker length
= 0.35435 m/s
Peak speed of the output pin at the rocker length c (a kinematic proxy for a cylinder only if the cylinder acts perpendicular to the rocker at that radius; the angular velocity is $\omega_{out,pk}$)
[50] $peak_{accel}$ = max $|a_{pin}|$ [4](cosines))
= $\sqrt{(\alpha_4^2 + \omega_4^4)}\cdot c$
= 3.1625 m/s²
Peak acceleration of the output pin
[51] $follower_{force}$ = max $|(T_L + J\cdot\alpha_4)/c|$ [5](work)
= tangential force at the output pin
= 43.476 N
Peak force perpendicular to the rocker at the output pin, the drive force of the output
[52] $F_{pin,pk}$ = max $|(T_L + J\cdot\alpha_4)/(c\cdot\sin\mu)|$ [5](work)
= force in the coupler = the crank-pin force
= 53.69 N
Peak force in the coupler, which is also the force at the crank pin and the pin loads on the bearings
[53] $T_{rms,in}$ = $\sqrt{(\text{mean } T_{in}^2)}$ [5](work)
= Simpson over the scan
= 1.3744 N·m
RMS input torque over the swept range

[54] $P_{in,pk} = \omega \cdot \max |T_{in}|$ [5](work)
= 6.2832 rad/s × 2.2654 N·m
= 0.014234 kW
Peak input power (the scale of the mean power)
[55] $P_{mean} = \omega \cdot \text{mean}(T_{in})$ [5](work)
= Simpson over the scan
= 0.0076496 kW
Mean input power over the swept range (negative: the mechanism returns energy)
[56] $u_{mu} = \mu_{req} / \mu_{min}$ [6](angle)
= 40° / 48.441°
= 0.82575
Transmission angle: required minimum against the smallest over the sweep

5. Results

SYMBOL	VALUE	DESCRIPTION
$input_{torque}$	2.265 N·m	Peak input (crank) torque by virtual work
$output_{motion}$	0.3543 m/s	Peak speed of the output pin at the rocker length c (a kinematic proxy for a cylinder only if the cylinder acts perpendicular to the rocker at that radius; the angular velocity is $\omega_{out,pk}$)
$peak_{accel}$	3.162 m/s²	Peak acceleration of the output pin
$follower_{force}$	43.48 N	Peak force perpendicular to the rocker at the output pin, the drive force of the output
$F_{pin,pk}$	53.69 N	Peak force in the coupler, which is also the force at the crank pin and the pin loads on the bearings
$T_{rms,in}$	1.374 N·m	RMS input torque over the swept range
n_{in}	60 rpm	Input (crank) speed, passed through
f_{in}	1 Hz	Input (crank) frequency, for a spring surge check
P_{mean}	0.007650 kW	Mean input power over the swept range (negative: the mechanism returns energy)
grashof	"Grashof crank-rocker, input link shortest"	Grashof classification
$S_{plus,L}$	195 mm	Shortest plus longest link
$P_{plus,Q}$	240 mm	Sum of the other two links
$theta2_{sw}$	[0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, … (73 items)] deg	Table: crank angle
$theta3_{sw}$	[45.45, 43.305, 41.183, 39.119, 37.143, 35.277, 33.533, 31.922, 30.446, 29.103, 27.892, 26.806, … (73 items)] deg	Table: coupler angle

SYMBOL	VALUE	DESCRIPTION
$theta4_{sw}$	[93.891, 91.886, 90.182, 88.806, 87.774, 87.09, 86.749, 86.738, 87.038, 87.625, 88.475, 89.561, … (73 items)] deg	Table: rocker angle
$omega3_{sw}$	[-2.6928, -2.6892, -2.6368, -2.5435, -2.4183, -2.2706, -2.1088, -1.9402, -1.7705, -1.6037, -1.4425, -1.2886, … (73 items)] rad/s	Table: coupler angular velocity
$omega4_{sw}$	[-2.6928, -2.3371, -1.9398, -1.5152, -1.0781, -0.64174, -0.21757, 0.18575, 0.56219, 0.90807, 1.2216, 1.5026, … (73 items)] rad/s	Table: rocker angular velocity
$alpha3_{sw}$	[-1.6439, 2.0963, 5.3499, 7.9793, 9.9336, 11.238, 11.97, 12.238, 12.157, 11.832, 11.354, 10.795, … (73 items)] rad/s²	Table: coupler angular acceleration
$alpha4_{sw}$	[23.794, 27.273, 29.765, 31.194, 31.596, 31.098, 29.878, 28.129, 26.035, 23.751, 21.398, 19.064, … (73 items)] rad/s²	Table: rocker angular acceleration
μ_{sw}	[48.441, 48.581, 48.999, 49.687, 50.631, 51.814, 53.216, 54.816, 56.592, 58.522, 60.583, 62.755, … (73 items)] deg	Table: transmission angle

SYMBOL	VALUE	DESCRIPTION
$T_{in,sw}$	[1.5919, 1.3661, 1.1246, 0.87436, 0.62126, 0.37043, 0.1261, 0.12823, 0.38586, 0.61929, 0.82764, 1.0113, … (73 items)] N·m	Table: input torque by virtual work
$theta3_0$	45.45°	Coupler angle at the first input angle
$theta4_0$	93.89°	Rocker angle at the first input angle
$omega3_0$	-2.693 rad/s	Coupler angular velocity at the first input angle
$omega4_0$	-2.693 rad/s	Rocker angular velocity at the first input angle
$alpha3_0$	-1.644 rad/s²	Coupler angular acceleration at the first input angle
$alpha4_0$	23.79 rad/s²	Rocker angular acceleration at the first input angle
μ_0	48.44°	Transmission angle at the first input angle
$T_{in,0}$	1.592 N·m	Input torque at the first input angle
$theta4_{alt}$	266.11°	Rocker angle on the other circuit at the first input angle
$theta4_{min}$	86.70°	Smallest rocker angle
$theta4_{max}$	141.49°	Largest rocker angle
swing	54.79°	Rocker swing over the sweep
μ_{min}	48.44°	Smallest transmission angle over the sweep
res_{max}	below 1e-10 mm	Largest loop-closure residual
$res_{vel,max}$	below 1e-12	Largest velocity-loop residual, relative to the size of its terms
$toggle_{angles}$	"none: coupler and rocker are never collinear"	Toggle crank angles (coupler and rocker collinear, μ = 0): the limits of the crank's assemblable range
$limit_{positions}$	"θ_2 = 32.66° with θ_4 = 86.704°; θ_2 = 220.95° with θ_4 = 141.49°"	Crank and coupler collinear (the rocker's dead-centre positions)
$omega3_{pk}$	2.698 rad/s	Peak coupler angular velocity
$\omega_{out,pk}$	3.543 rad/s	Peak output (rocker) angular velocity
$alpha3_{pk}$	22.72 rad/s²	Peak coupler angular acceleration
$\alpha_{out,pk}$	31.60 rad/s²	Peak output (rocker) angular acceleration
u_{mu}	0.83	Required against smallest transmission angle (governing)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Required transmission angle against the smallest transmission angle	40°	48.44°	0.83	[6](angle)
PASS	Loop-closure residual against its limit	max residual below 1e-10 mm	< 1e-9 mm	—	[7]
PASS	Velocity-loop residual against its limit	max residual below 1e-12	< 1e-8	—	[7]

7. Applicability

None.

8. Verification

Template mech.mechanism@1.3.1 Content hash 7366a85fb532ee70 Engine calc-engine/1.0.0
Known-answer cases passed 16/16 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. Norton, Design of Machinery.
2. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.
3. Grashof.
4. Vector-loop closed form (Freudenstein / law of.
5. Virtual.
6. Transmission.
7. Self-consistency.